MILITARY SPECIFICATION

DIAPHRAGM, VAGINAL, RUBBER

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers contoured, rubber, vaginal diaphragms suitable for use with spermicidal, vaginal cream or jelly for conception control.

1.2 Classification. Diaphragms shall be furnished in the following sizes, as indicated below (see 6.1).

SIZE

1 - 70 millimeters (mm.)
2 - 75 mm.
3 - 80 mm.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

GG-I-526	Instruments, Dental and Surgical, General Specifications.
MMM-A-250	Adhesive, Water-Resistant (for Sealing Fiberboard boxes).
PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Boxes, Set-Up.
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper (for Carton Sealing).

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STANDARDS

Federal

Fed. Test Method Std. No. 601 Rubber: Sampling and Testing.

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM Designation A228 Music Steel Spring Wire

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103).

U. S. DEPARTMENT OF HEALTH, EDUCATION AND WELFARE

Federal Food, Drug and Cosmetic Act (FDA) and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, D. C. 20204).

3. REQUIREMENTS

3.1 Material.

3.1.1 Rubber. The dome or cup portion and the bead or rim of the diaphragm shall be fabricated from natural rubber latex. The rubber compound shall be free from any ingredient which may cause any irritating effect upon the body.

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3.1.2 Metal. The diaphragm flat spring shall be fabricated from stainless steel in accordance with the requirements of GG-I-526 for class 3, type 304 material. The diaphragm coil spring shall be fabricated from tin-coated music steel spring wire meeting the requirements of ASTM Designation A228.

3.2 Style, design, and construction. The vaginal diaphragm shall consist of a soft-textured rubber dome or cup and an integral, peripheral rim. The rim shall be formed by a molded, annular rubber bead which shall encase a flexible, distortion-free dual spring. The dual spring shall consist of a flat strip of stainless steel, flat wound to form three complete turns, within a tin-coated steel coiled spring. The spring shall be sufficiently rigid to hold the rim in a flat, circular position so that the cup portion is held in its extended position and is completely flexible throughout. The spring shall also permit the rim to be flexed. Whenever two opposite sides of the rim are squeezed together, the rim shall form an arc and shall resist saddling, distortion, twisting, and breaking and the flexible cup portion shall become draped to form a pocket. When tested as specified in 4.4.1, the diaphragm shall be freely flexible and shall return to its original shape without evidence of saddling, deformation, or distortion. When tested as specified in 4.4.5, the dual spring shall show no evidence of breakage or distortion and shall return to its original shape.

3.2.1 Dimensions. The dimensions of the diaphragm shall be in accordance with Table I. The thickness of the dome or cup shall be measured as specified in 4.4.2.

Table I. Dimensions

Characteristic	Measurement
Overall diameter of diaphragm:	
Size 1	70 ± 2 mm.
Size 2	75 ± 2 mm.
Size 3	80 ± 2 mm.
Inside diameter of diaphragm (excluding rim)	
Size 1	2.371 ± 0.015 inch (in.)
Size 2	2.568 ± 0.015 in.
Size 3	2.765 ± 0.015 in.
Height of dome	
Size 1	0.969 ± 0.031 in.
Size 2	1.031 ± 0.031 in.
Size 3	1.094 ± 0.031 in.

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Table I. Dimensions (Cont'd.)

Thickness of dome (exclusive of normal dimple at top of dome)	0.008 to 0.019 in.
Diameter of rim	0.215 ± 0.01 in.
Overall diameter of dual spring	0.150 ± 0.002 in.
Diameter of coil spring wire	0.030 ± 0.0005 in.
Coils per inch	17 ± 1
Width of flat spring	0.080 ± 0.002 in.
Thickness of flat spring	0.005 ± 0.0005 in.

3.3 Physical properties. The physical properties of the rubber shall be as specified in Table II, when tested as specified in 4.4.3.

Table II. Physical Properties

Characteristic	Requirement
Tensile strength in pounds per square inch (psi)	2,200 minimum
Ultimate elongation in percentage	650 minimum
Accelerated Aging	
Tensile strength, psi	1,650 minimum
Ultimate elongation, percentage	485 minimum
Specific gravity	0.95 + .02 - .05

3.4 Finish. The diaphragms shall have a uniform smooth finish free from holes, visible surface or embedded particles, tackiness, bloom, bleeding, and other imperfections.

3.5 Freedom from pinholes. The diaphragm shall be supplied free of pinholes and thin or weak spots when tested as specified in 4.4.6.

3.6 Water leakage. The diaphragm shall not leak when tested as specified in 4.4.4. The diaphragms shall be tested for water leakage after the pinhole test.

3.7 Identification marking. The diaphragm shall be legibly and permanently marked with the manufacturer's name or registered trademark and the size designation in mm.

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3.8 Instructions. Each diaphragm shall be provided with a detailed instruction booklet containing the following minimum information: description and function of the diaphragm; when the diaphragm should be inserted and removed; techniques for correct insertion and removal with appropriate anatomy illustrations; and proper care and replacement of the diaphragm.

3.9 Workmanship. Workmanship shall be first class throughout. The diaphragms shall be free from defects which detract from their appearance or impair their serviceability. Diaphragms shall be packed under sanitary conditions.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsiblity for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality supplied by the manufacturer of the rubber or the metal may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the FSN, the item identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, the individual test results, and the specification requirements.

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4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and as indicated in Table III. The unit of product for examination purposes shall be one diaphragm.

Table III. Sampling for Examination

Defect category	Inspection level	AQL (percent defective)
Major	II	1.0
Minor	II	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as indicated in Table IV.

Table IV. Sampling for Tests

Component (unit of product)	Characteristic	Requirement	Test procedure	Inspection level	AQL (percent defective)
1-Diaphragm	Flexibility	3.2	4.4.1	S-2	*
1-Diaphragm	Dimensions-	3.2.1	---	S-2	*
1-Diaphragm	Thickness	3.2.1	4.4.2	S-2	*
1-Diaphragm	Tensile strength	3.3	4.4.3	S-2	*
1-Diaphragm	Ultimate elongation	3.3	4.4.3	S-2	*
1-Diaphragm	Accelerated aging	3.3	4.4.3	S-2	*
1-Diaphragm	Specific gravity	3.3	4.4.3	S-2	*
1-Diaphragm	Freedom from pinholes	3.5	4.4.6	**	*
1-Spring	Coiling	3.2	4.4.5	S-2	*
1-Diaphragm	Water leakage	3.6	4.4.4	S-2	*

* Acceptance number shall be zero for all sample sizes.
** The pinhole test shall be performed on 100 percent basis (no sampling).

4.2.3 Sample unit usage. Where possible, the same sample unit shall be used for the determination of two or more test characteristics.

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4.3 Examination. The diaphragms shall be examined to determine compliance with specification requirements. Examination shall be conducted in accordance with MIL-STD-105 and the following classification of defects. Visual examination shall include inflation of the diaphragm to be examined. This inflation shall be limited to 350 ± 50 percent elongation on any part of the diaphragm. Examination shall include, but shall not be limited to, the defects listed in Table V.

Table V. Classification of Defects

Categories		Defects
Critical		None defined.
Major:		
	101	Diaphragm not symmetrical in shape or not free of deformation.
	102	Spring not completely or uniformly molded into rim.
	103	Spring bent, broken, twisted, or distorted.
	104	Rim of diaphragm not smooth, soft or free of rough edges or discoloration.
	105	Rim of diaphragm not free of cuts, tears, or slits.
	106	Diaphragm not free of open blisters.
	107	Diaphragm not free of embedded, grit-like particles.
	108	Diaphragm not free of tackiness, blooming, or bleeding.
	109	Diaphragm not free of dirt, grease, or foreign matter.
	110	Instruction booklet missing, incomplete, or illegible.
Minor:		
	201	Diaphragm not free of air bubbles, mica specks or other surface blemishes.
	202	Dome of diaphragm not free of discoloration.
	203	Identification marking not present, complete, correct, legible, or permanent.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements. All sample or test specimens intended for accelerated aging tests shall be protected from any unnecessary exposure to strong light prior to the aging. The samples and specimens shall be kept in light-tight containers, except while being inspected, prepared or measured.

4.4.1 Flexibility tests. The diaphragm shall be squeezed together at two exactly opposite sides on the rim, held for 5 seconds, and released. Three separate consecutive tests shall be performed for each diaphragm.

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4.4.2 Dimensional test. Thickness shall be measured in accordance with the test method of Fed. Test Method Standard No. 601, indicated in Table VI.

4.4.3 Physical properties tests. Tests to determine physical properties shall be conducted in accordance with the test methods of Fed. Test Method Standard No. 601, indicated in Table VI.

Table VI. Test Methods for Physical Properties

Requirement	Test Method No. 601
Thickness	2011 or 2031
Tensile strength	4111 *
Ultimate elongation	4121 *
Accelerated aging	7111 *
Specific gravity	14011 or 14021

* Use ring specimen.

4.4.4 Leakage test. The rim of the diaphragm shall be fitted over the mouth of a glass filter funnel having a beaded edge, such as a perforated Buchner funnel of appropriate outside diameter. The diaphragm shall be secured to the mouth by means of a rubber band. The funnel with the attached diaphragm shall be inverted and mounted on a ring stand. A two-foot length of glass tubing shall be attached to the stem of the funnel by means of a short piece of rubber tubing. The assembly shall then be filled with water through the open end of the glass tubing to a height of 20 inches measured from the mouth edge of the funnel. After 10 minutes the diaphragm shall be examined for any evidence of leakage.

4.4.5 Spring coiling test. The dual spring shall be wrapped around a 3/4-inch dowel, held for 5 seconds, and released.

4.4.6 Pinhole test. The dome of the diaphragm shall be stretched to twice its normal size and examined over the light transmitted by a 25-watt bulb.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One diaphragm, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in Table VII shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

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Table VII. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	10 units	200 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Each diaphragm shall be packaged in a rigid plastic case of appropriate size and design in such a manner as not to damage the diaphragm. The case containing the diaphragm shall be further packaged in a box of appropriate size and design constructed in accordance with PPP-B-566, or PPP-B-676, except commercial colors will be acceptable. Closure shall be adequate to prevent accidental opening.

5.1.3.2 Intermediate package. Intermediate package shall be a box of appropriate size and design constructed in accordance with PPP-B-636, type CF, class weather-resistant. Boxes shall be closed by taping the top and bottom flaps with 2 inch wide tape conforming to PPP-T-76. Closing of bottom flap with adhesive conforming to MMM-A-250 or metal stitches is permitted.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3. Closure and strapping shall be as specified in the appendix of the applicable box specification.

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5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packaged in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Unit package. Each unit package shall be marked in accordance with MIL-STD-129 and the requirements of the Federal Food, Drug, and Cosmetic Act. In addition, the contract or purchase order number and name of the contractor, the date of manufacture (in lieu of date packed) shall be shown. Instructions shall be inserted in each unit package as specified in 3.7.

5.3.2 Intermediate packages. Each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown. Date of manufacture shall be shown in lieu of date packed.

5.3.3 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. Date of manufacture shall be shown in lieu of date packed.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Number, title and date of this specification.
(b) Applicable size (see 1.2).
(c) Applicable levels of packing and packaging (see 5.1 and 5.2).
(d) Federal stock number.

6.2 This specification covers the following items appearing in the Federal Supply Catalog:

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Federal Stock Number	Item Identification
6515-935-1025	DIAPHRAGM, VAGINAL, Rubber, Contoured, 70 mm.
6515-935-1026	DIAPHRAGM, VAGINAL, Rubber, Contoured, 75 mm.
6515-935-1028	DIAPHRAGM, VAGINAL, Rubber, Contoured, 80 mm.

It does not include all types or sizes as indicated by the title, nor all types commercially available.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6515-1494

Review activities:
Army - MD
Navy - MS
Air Force - 03

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document (s) or serve to amend contractual requirements.

SPECIFICATION
MIL-D-36691 FSN:

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 61, WHICH MAY BE USED.

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CAUTION

NOTICE TO BIDDERS/OFFERORS

DO NOT CONDITION OR BASE YOUR BID/OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID/OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

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